Chapter 6

Aerodynamics of Aircraft in Maneuver

This chapter will examine the aerodynamics of thin wings of arbitrary planform and of slender bodies in arbitrary translation and rotation. Quasi-steady flow will be assumed.

6.1 Aircraft Motion Definition

Chapter 9 will derive in detail the Earth and body axis systems used for describing aircraft motion. Here, a few of those key relations will be simply stated without derivation. Unless otherwise indicated, all vector components will be assumed to be in the *geometry axes* shown in Figure 6.1, which have x and z reversed from the body axes given in Chapter 9. The other axis systems will be discussed where appropriate.

6.1.1 Aircraft velocity and rotation

The aircraft motion is defined by the velocity $\mathbf{U}$ of its axis-origin point, and by its rotation rate $\mathbf{\Omega}$. Both are shown in Figure 6.1. These are defined relative to the Earth frame, and hence they are also the velocity and rotation rate of the aircraft relative to a still airmass.

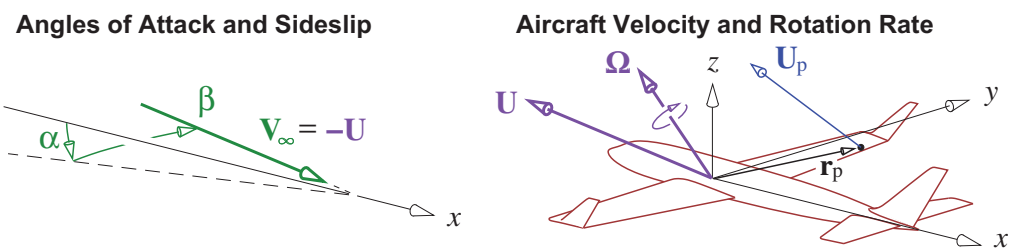

Figure 6.1: Aircraft velocity and rotation rate $\mathbf{U}, \mathbf{\Omega}$, and resulting velocity $\mathbf{U}_\mathrm{p}$ of body point $\mathbf{r}_\mathrm{p}$. Normalized velocity $\mathbf{U}/V_\infty$ is specified by the angles of attack and sideslip α, β. For computation, all vectors are specified via their components along the aircraft's xyz geometry axes.

The aerodynamic "freestream" velocity $\mathbf{V}_\infty$ is directly opposite to $\mathbf{U}$, and is conventionally specified by the two aerodynamic flow angles α and β, applied in that order as shown in Figure 6.1.

$$\mathbf{U} \;=\; \begin{Bmatrix} U_x \\ U_y \\ U_z \end{Bmatrix} \;=\; -\mathbf{V}_\infty \;=\; V_\infty \begin{Bmatrix} -\cos\alpha\,\cos\beta \\ \sin\beta \\ -\sin\alpha\,\cos\beta \end{Bmatrix} \tag{6.1}$$

$$V_\infty \;=\; \sqrt{U_x^2 + U_y^2 + U_z^2} \quad , \qquad \alpha \;=\; \arctan\frac{-U_z}{-U_x} \quad , \qquad \beta \;=\; \arctan\frac{U_y}{\sqrt{U_x^2 + U_z^2}} \tag{6.2}$$

Given these reciprocal relations, $\{V_\infty, \alpha, \beta\}$ and $\{U_x, U_y, U_z\}$ are equivalent alternative parameter sets. In practice, α, β are chosen as the independent parameters. These define the three components of the normalized aircraft velocity $\mathbf{U}/V_\infty$ via (6.1), which are needed to compute the aerodynamic forces and moments.

6.1.2 Body-point velocity

The Earth-frame velocity of any point $\mathbf{r}_\mathrm{p}$ fixed on the body is given by

$$\mathbf{U}_\mathrm{p} \;=\; \mathbf{U} + \mathbf{\Omega}\times\mathbf{r}_\mathrm{p} \tag{6.3}$$

as shown in Figure 6.1. If the airmass is still (without wind or gusts), then the apparent airmass velocity seen by this point is $-\mathbf{U}_\mathrm{p}$, which is in effect a "local freestream." This will be used to formulate flow-tangency boundary conditions in computational methods.

6.2 Axis Systems

In computational methods, the aircraft motion vectors $\mathbf{U}, \mathbf{\Omega}$, and the aerodynamic force and moment vectors $\mathbf{F}, \mathbf{M}$ are most easily specified or calculated in the same xyz axes which are used to specify the geometry itself, shown in Figure 6.1. To apply the results to aircraft performance, stability and control, and other related disciplines it is necessary to provide these vector quantities in other more relevant axes.

6.2.1 Stability axes

The drag, sideforce, and lift force components are most commonly defined in the *stability axes*, which are rotated from the geometry axes by only the angle of attack α (not by sideslip β), as shown in Figure 6.2.

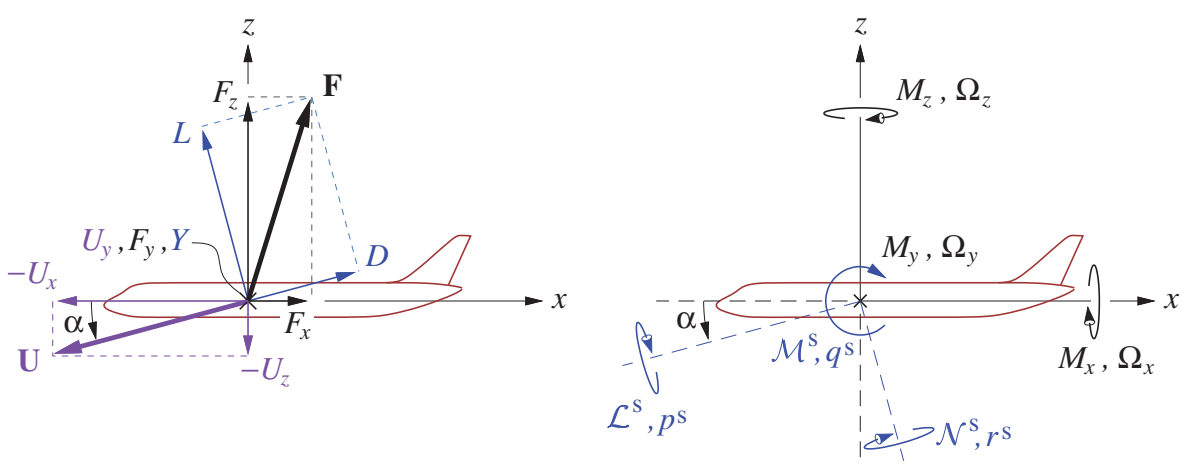

Figure 6.2: Orientation of stability-axis (freestream-aligned) aerodynamic force components D, Y, L, moment components $\mathcal{L}^\mathrm{s}, \mathcal{M}^\mathrm{s}, \mathcal{N}^\mathrm{s}$, and rotation rate components $p^\mathrm{s}, q^\mathrm{s}, r^\mathrm{s}$.

All the standard vector components in stability axes are defined from their components in geometry axes using the $\bar{\bar{\mathbf{T}}}^\mathrm{s}$ rotation matrix.

$$\bar{\bar{\mathbf{T}}}^\mathrm{s} \;=\; \begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix} \tag{6.4}$$

$$\begin{Bmatrix} D \\ Y \\ L \end{Bmatrix} = \begin{bmatrix} \\ \bar{\bar{\mathbf{T}}}^\mathrm{s} \\ \\ \end{bmatrix} \begin{Bmatrix} F_x \\ F_y \\ F_z \end{Bmatrix} \;,\quad \begin{Bmatrix} \mathcal{L}^\mathrm{s} \\ \mathcal{M}^\mathrm{s} \\ \mathcal{N}^\mathrm{s} \end{Bmatrix} = \begin{bmatrix} \\ \bar{\bar{\mathbf{T}}}^\mathrm{s} \\ \\ \end{bmatrix} \begin{Bmatrix} -M_x \\ M_y \\ -M_z \end{Bmatrix} \;,\quad \begin{Bmatrix} p^\mathrm{s} \\ q^\mathrm{s} \\ r^\mathrm{s} \end{Bmatrix} = \begin{bmatrix} \\ \bar{\bar{\mathbf{T}}}^\mathrm{s} \\ \\ \end{bmatrix} \begin{Bmatrix} -\Omega_x \\ \Omega_y \\ -\Omega_z \end{Bmatrix} \tag{6.5}$$

Note that the moment and rotation-rate components, $\mathcal{L}^\mathrm{s}, \mathcal{N}^\mathrm{s}$ and $p^\mathrm{s}, r^\mathrm{s}$, have reverse signs compared to the force components D, L. In effect, the stability axes used for the moments and rates are rotated by $180°$ about the y axis relative to the stability axes used for the forces. Note also that the $\bar{\bar{\mathbf{T}}}^\mathrm{s}$ matrix leaves all the vector y components unchanged. Hence we have $Y = F_y$, $\mathcal{M}^\mathrm{s} = M_y$, $q^\mathrm{s} = \Omega_y$.

6.2.2 Wind axes

The stability axes are not quite appropriate when examining the drag of an aircraft in sideslipping flight with $\beta \neq 0$, since the drag D as defined by the $\bar{\bar{\mathbf{T}}}^{\mathrm{s}}$ matrix in (6.5) is <u>not</u> the true streamwise drag force. In this situation we can invoke the *wind axes*, which are implemented by the rotation matrix $\bar{\bar{\mathbf{T}}}^{\mathrm{w}}$, which consists of α and β rotations, applied in that order as shown in Figure 6.1.

$$\begin{Bmatrix} D \\ Y \\ L \end{Bmatrix} = \begin{bmatrix} & & \\ & \bar{\bar{\mathbf{T}}}^{\mathrm{w}} & \\ & & \end{bmatrix} \begin{Bmatrix} F_x \\ F_y \\ F_z \end{Bmatrix} \tag{6.6}$$

$$\bar{\bar{\mathbf{T}}}^{\mathrm{w}} = \begin{bmatrix} \cos\beta & -\sin\beta & 0 \\ \sin\beta & \cos\beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix} = \begin{bmatrix} \cos\beta\cos\alpha & -\sin\beta & \cos\beta\sin\alpha \\ \sin\beta\cos\alpha & \cos\beta & \sin\beta\sin\alpha \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix} \tag{6.7}$$

Note that the drag as defined by the $\bar{\bar{\mathbf{T}}}^{\mathrm{w}}$ matrix in (6.6) is exactly equivalent to the dot product of the total force and the unit freestream.

$$D = \mathbf{F} \cdot \mathbf{V}_\infty / V_\infty \tag{6.8}$$

Furthermore, the sideforce Y produced by $\bar{\bar{\mathbf{T}}}^{\mathrm{w}}$ in (6.6) is almost the same as that produced by $\bar{\bar{\mathbf{T}}}^{\mathrm{s}}$ in (6.5), and the lift L is identical. Because the simple relation (6.8) is available to define the exact D when needed, and the exact Y is of relatively little importance, wind axes see little use in practice.

6.3 Non-Dimensionalization and Parameterization

6.3.1 Dimensionless variables

Aerodynamic characteristics are almost invariably defined, described, or provided in terms of the following dimensionless force coefficients, moment coefficients, and rotation rates. The latter two are commonly used in either the body or the stability axes.

$$\begin{aligned} C_D &= \frac{D}{q_\infty S_{\mathrm{ref}}} & C_\ell &= \frac{\mathcal{L}}{q_\infty S_{\mathrm{ref}}\, b_{\mathrm{ref}}} & \bar{p} &= \frac{p\, b_{\mathrm{ref}}}{2V_\infty} \\ C_Y &= \frac{Y}{q_\infty S_{\mathrm{ref}}} & C_m &= \frac{\mathcal{M}}{q_\infty S_{\mathrm{ref}}\, c_{\mathrm{ref}}} & \bar{q} &= \frac{q\, c_{\mathrm{ref}}}{2V_\infty} \\ C_L &= \frac{L}{q_\infty S_{\mathrm{ref}}} & C_n &= \frac{\mathcal{N}}{q_\infty S_{\mathrm{ref}}\, b_{\mathrm{ref}}} & \bar{r} &= \frac{r\, b_{\mathrm{ref}}}{2V_\infty} \end{aligned} \tag{6.9}$$

All the reference quantities are arbitrary. The traditional choice for the reference area S_{ref} is the projected wing area S. This typically includes any hidden "carry-through" wing area inside a fuselage, but may exclude root fairings or fillets. The choice for b_{ref} is the actual projected wingspan, which may or may not include tip devices such as winglets. The traditional choice for c_{ref} is the wing's *mean aerodynamic chord*,

$$c_{\mathrm{ref}} = c_{\mathrm{mac}} \equiv \frac{1}{S} \int_{-b/2}^{b/2} c_{(y)}{}^2 \, \mathrm{d}y \tag{6.10}$$

which is in effect a root-mean-square chord. Workable alternatives are the average chord $c_{\mathrm{ref}} = S/b$, or simply the root chord $c_{\mathrm{ref}} = c_{(0)}$.

Theoretically, additional important parameters are the dimensionless flow-angle rates.

$$\bar{\dot{\alpha}} = \frac{\dot{\alpha} c_{\mathrm{ref}}}{2V_\infty} \qquad \bar{\dot{\beta}} = \frac{\dot{\beta} b_{\mathrm{ref}}}{2V_\infty} \tag{6.11}$$

This $\bar{\dot{\alpha}}$ quantifies the strength and influence of the wing's *shed vorticity*, which is present in unsteady airfoil flows and is discussed in more detail in Section 7.4.2. In brief, $\bar{\dot{\alpha}}$ determines the time delay in the wing's downwash seen by the horizontal tail, and therefore during pitching maneuvers it influences the time evolution of the overall C_m, and to a lesser extent of the C_L also. In most aircraft $\bar{\dot{\beta}}$ has relatively little influence and is usually ignored, although it may be significant for unusual aircraft configurations.

6.3.2 Quasi-steady force and moment parameterization

Dimensional analysis indicates that all the force and moment coefficients have the following parametric dependence for steady flows. The same dependencies also approximately hold for quasi-steady flows such as an aircraft in slow maneuver.

$$\begin{aligned}
C_D &= C_D(\alpha, \beta, \bar{p}, \bar{q}, \bar{r}, \bar{\dot{\alpha}}, \bar{\dot{\beta}}, \delta_T, \delta_a, \delta_e, \delta_r, M_\infty, Re_\infty) \\
C_Y &= C_Y(\alpha, \beta, \bar{p}, \bar{q}, \bar{r}, \bar{\dot{\alpha}}, \bar{\dot{\beta}}, \delta_T, \delta_a, \delta_e, \delta_r, M_\infty, Re_\infty) \\
&\vdots \\
C_n &= C_n(\alpha, \beta, \bar{p}, \bar{q}, \bar{r}, \bar{\dot{\alpha}}, \bar{\dot{\beta}}, \delta_T, \delta_a, \delta_e, \delta_r, M_\infty, Re_\infty)
\end{aligned}$$

The parameters $\delta_T, \delta_a, \delta_e, \delta_r$ are throttle, aileron, elevator, rudder *control parameters* (there may be more). These represent thrust settings or control-surface deflection angles, which influence the overall force and moment on the aircraft.

Figure 6.3 shows typical flows over the entire possible α, β range, most of which involve large-scale separation and flow reversal. The result is that the force and moment coefficients have complicated dependencies on the operating parameters, as indicated in the sample $C_{L(\alpha)}$ and $C_{n(\beta)}$ line plots in Figure 6.3.

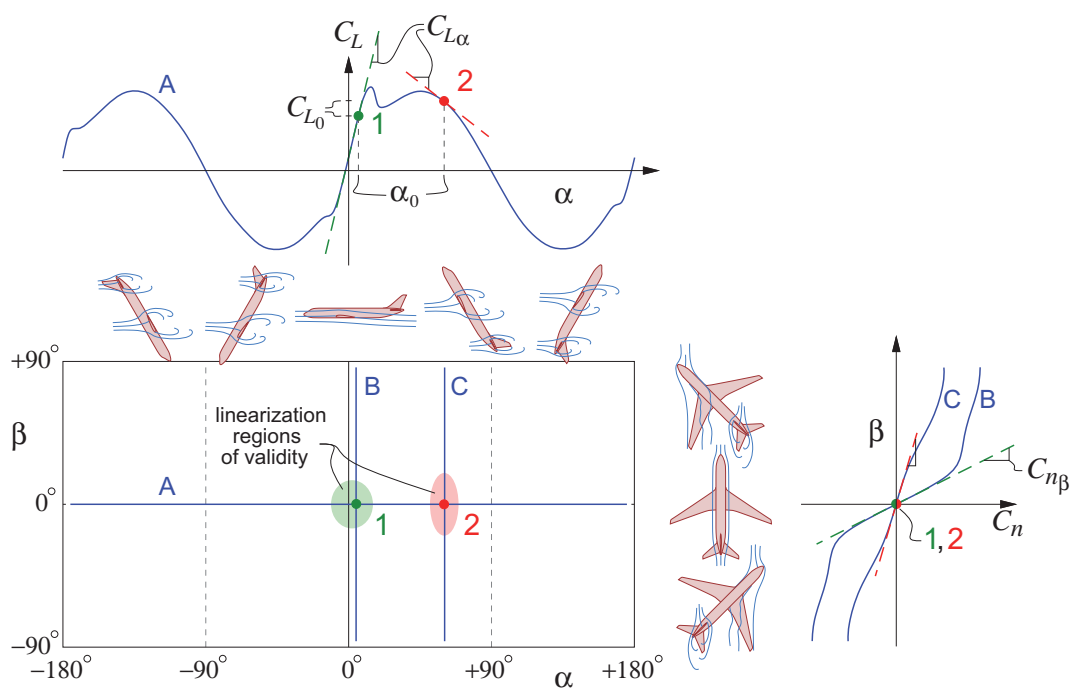

Figure 6.3: Aircraft α, β flow parameter space (remaining parameters $\bar{p}, \bar{q}, \bar{r} \ldots$ not shown). $C_{L(\alpha)}$ variation is shown for parameter slice A, and $C_{n(\beta)}$ variation is shown for slices B, C. The linearized force or moment representations, indicated by dashed lines, are valid in a sufficiently small region near the chosen baseline operating point. Two baseline points 1, 2 at two different α_0 values are indicated, with the stability derivative C_{L_α} changing dramatically between the two points.

The overall parameter space is enormously larger than the two-axis slice shown in Figure 6.3, since there are additional axes for $\bar{p}, \bar{q}, \bar{r}, \bar{\dot{\alpha}}, \bar{\dot{\beta}}, \delta_T \ldots, M_\infty, Re_\infty$. For many applications, such as mission performance estimates, stability and control analysis and design, etc., the force and moment coefficients only need to be defined within a small region of the parameter space, described by small deviations about some *operating point* or *trim state*, denoted by the $()_0$ subscript. Here the force and moment coefficient functions can be approximated by their <u>linearized</u> forms, or equivalently their first-order Taylor series approximations.

$$\begin{aligned} C_L &\simeq C_{L_0} + C_{L_\alpha}\,\Delta\alpha + C_{L_q}\,\Delta\bar{q} + C_{L_{\dot{\alpha}}}\,\Delta\bar{\dot{\alpha}} + C_{L_{\delta_e}}\,\Delta\delta_e && (6.12)\\ C_n &\simeq C_{n_0} + C_{n_\beta}\,\Delta\beta + C_{n_p}\,\Delta\bar{p} + C_{n_r}\,\Delta\bar{r} + C_{n_{\delta_r}}\,\Delta\delta_r && (6.13)\\ &\vdots \end{aligned}$$

where

$$\begin{aligned} C_{L_0} &= C_{L(\alpha_0,\beta_0,\ldots)} & C_{L_\alpha} &= \partial C_L/\partial\alpha\,{}_{(\alpha_0,\beta_0,\ldots)} & C_{L_q} &= \partial C_L/\partial\bar{q}\,{}_{(\alpha_0,\beta_0,\ldots)} & \cdots\\ C_{n_0} &= C_{n(\alpha_0,\beta_0,\ldots)} & C_{n_\beta} &= \partial C_n/\partial\beta\,{}_{(\alpha_0,\beta_0,\ldots)} & C_{n_p} &= \partial C_n/\partial\bar{p}\,{}_{(\alpha_0,\beta_0,\ldots)} & \cdots \qquad (6.14)\\ &\vdots & &\vdots & &\vdots \end{aligned}$$

The series variables are the aerodynamic parameter perturbations from the trim state.

$$\Delta\alpha = \alpha - \alpha_0 \qquad \Delta\beta = \beta - \beta_0 \qquad \Delta\bar{p} = \bar{p} - \bar{p}_0 \qquad \ldots \tag{6.15}$$

The series coefficients $C_{L_\alpha}, C_{L_q} \ldots$ are *stability derivatives*, and $C_{L_{\delta_e}}, C_{n_{\delta_r}} \ldots$ are *control derivatives*. These play a crucial role in aircraft flight dynamics and stability and control, as outlined in Chapter 9. Note that these coefficients can substantially depend on the baseline trim state values, and some may have their signs reversed between different trim states, as for example C_{L_α} shown in Figure 6.3.

6.4 Lifting Surface Theory

Lifting surface theory is an extension of thin airfoil theory to 3D. It models the flow about the wings and tails of a general 3D aircraft configuration using vortex sheets $\boldsymbol{\gamma}_{(s,\ell)}$, or the equivalent normal-doublet sheets $\mu_{(s,\ell)}$. The objective is to represent the lift, sideforce, moments, and induced drag of the configuration using only the camber surface shapes, with the volume effects of the various components being ignored.

6.4.1 Vortex/doublet sheet geometry

The assumed geometry of the vortex or doublet sheets is shown in Figure 6.4. The sheets are assumed to be everywhere parallel to the x axis, with the camber-surface shapes of the actual geometry represented only by their normal vector distribution $\hat{\mathbf{n}}_{(s,\ell)}$. The sheet strengths $\boldsymbol{\gamma}_{(s,\ell)}$ or $\mu_{(s,\ell)}$ are unknown only over the extent of the actual surface. On the trailing wake portions of the sheets, the strengths are constant in x, and equal to their trailing-edge values.

$$\begin{aligned} \gamma_{(s)} &= \boldsymbol{\gamma}_{(s,\ell_{TE})}\cdot\hat{\mathbf{x}} && \text{(on wake)}\\ \text{or}\qquad \mu_{(s)} &= \mu_{(s,\ell_{TE})} && \text{(on wake)} \end{aligned} \tag{6.16}$$

These are also the sheet strengths in the Trefftz plane, as shown in Figure 6.4.

6.4.2 Lifting-surface problem formulation

The perturbation velocity field of the vortex sheet distribution is given by the superposition integral (2.16),

$$\mathbf{V}_{\gamma(\mathbf{r})} = \frac{1}{4\pi}\iint \boldsymbol{\gamma}_{(s,\ell)}\times\frac{\mathbf{r}-\mathbf{r}'}{|\,\mathbf{r}-\mathbf{r}'|^3}\,\mathrm{d}s\,\mathrm{d}\ell \tag{6.17}$$

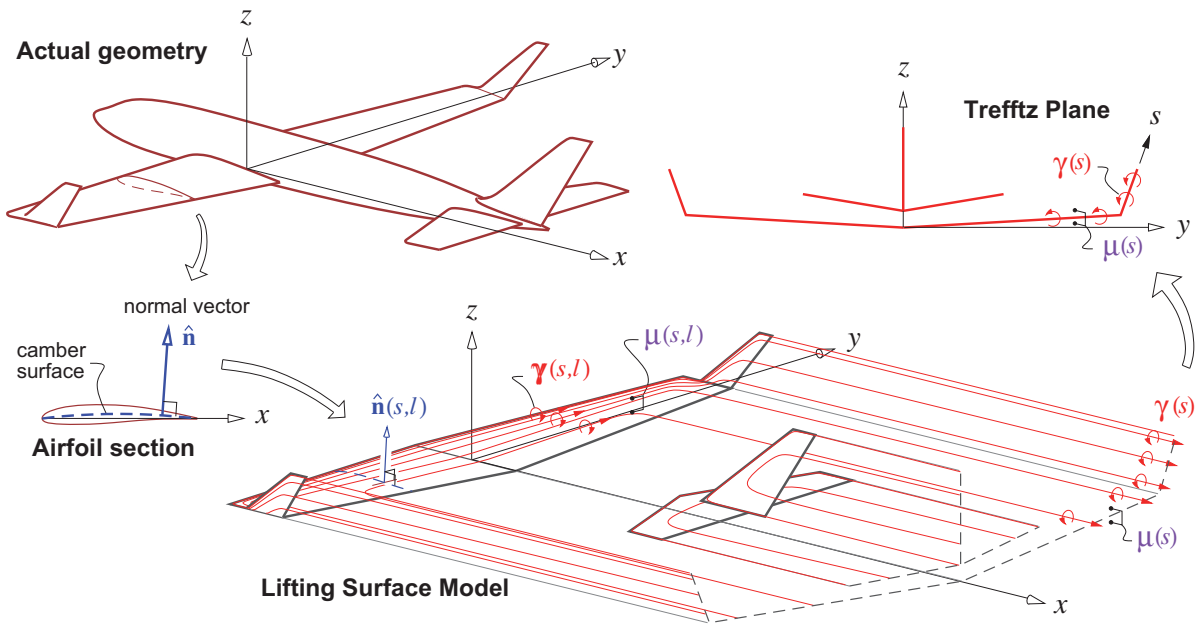

Figure 6.4: 3D configuration represented using a lifting-surface model. The model consists of vortex sheets with strength $\boldsymbol{\gamma}_{(s,\ell)}$ on the surfaces, and $\gamma_{(s)}\,\hat{\mathbf{x}}$ on the trailing wakes and in the Trefftz plane.

where $\mathbf{r}'_{(s,\ell)}$ is the assumed vortex sheet geometry. The integral is over both the surface and wake vortex sheets. With an airmass which is still in the Earth frame, the total fluid velocity observed by a point $\mathbf{r}$ fixed in the body frame is then obtained by subtracting that point's velocity $\mathbf{U}_{\mathrm{p}}$ as given by (6.3).

$$\mathbf{V}_{(\mathbf{r})} \;=\; \mathbf{V}_{\gamma} \;-\; (\mathbf{U} + \boldsymbol{\Omega}\times\mathbf{r}) \tag{6.18}$$

The flow-tangency boundary condition is then

$$\mathbf{V}_{(\mathbf{r})}\cdot\hat{\mathbf{n}} \;=\; \Big[\,\mathbf{V}_{\gamma} \;-\; (\mathbf{U} + \boldsymbol{\Omega}\times\mathbf{r})\Big]\cdot\hat{\mathbf{n}} \;=\; 0 \tag{6.19}$$

for each $\mathbf{r}_{(s,\ell)}$ surface point. The Kutta condition

$$\boldsymbol{\gamma}_{(s,\ell_{TE})}\times\hat{\mathbf{x}} \;=\; \mathbf{0} \tag{6.20}$$

is also applied all along the trailing edge of each surface. Equations (6.17), (6.19), (6.20) together constitute an integral-equation problem for the unknown $\boldsymbol{\gamma}_{(s,\ell)}$ distribution. If instead $\mu_{(s,\ell)}$ is chosen as the unknown variable, then $\boldsymbol{\gamma} = \hat{\mathbf{n}}\times\widetilde{\nabla}\mu$ would be substituted into all the above expressions. This lifting-surface problem can be solved by the Vortex Lattice Method, described later in Section 6.5.

6.4.3 Near-field loads

Once the surface vortex sheet strength $\boldsymbol{\gamma}_{(s,\ell)}$ distribution is known from the solution of the lifting-surface problem as described above, the resulting pressure jump or equivalently the loading can be determined using the Bernoulli equation as follows.

$$\Delta p \;\equiv\; p_l \;-\; p_u \;=\; \tfrac{1}{2}\rho\,\Delta(\mathbf{V}\cdot\mathbf{V}) \;=\; \rho\,\mathbf{V}_a\cdot\Delta\mathbf{V} \tag{6.21}$$

$$\begin{aligned}\Delta\mathbf{V} \;&\equiv\; \mathbf{V}_u - \mathbf{V}_l \;=\; \boldsymbol{\gamma}\times\hat{\mathbf{n}}\\ \mathbf{V}_a \;&\equiv\; \tfrac{1}{2}(\mathbf{V}_u+\mathbf{V}_l) \;=\; (\mathbf{V}_{\gamma})_a \;-\; (\mathbf{U} + \boldsymbol{\Omega}\times\mathbf{r})\end{aligned}$$

The $()_u$ and $()_l$ subscripts denote the upper and lower sides of the surface, as also used in Section 5.4 for a wake. Here $(\mathbf{V}_\gamma)_a$ is the velocity of the entire vortex sheet configuration, averaged between two field points on the two sides of the sheet. An equivalent version of the loading expression (6.21) is

$$\Delta p\, \hat{\mathbf{n}} \;=\; \rho\, \mathbf{V}_a \times \boldsymbol{\gamma} \tag{6.22}$$

which in effect is a local Kutta-Joukowsky relation.

The lifting surface approximation largely neglects the details of the flow in the leading edge region, which consequently requires a special treatment in the force calculations. Figure 6.5 shows the actual $(p_\infty - p)\,\hat{\mathbf{n}}$ surface load vectors on 2D inviscid airfoils of different thicknesses. The strongly negative pressure distribution acting on the small leading radius is known as *leading edge suction*, and is the mechanism by which the pressure drag of the aft-pointing pressure forces over the rest of the airfoil are canceled. In 2D the cancellation is theoretically perfect, while in 3D the cancellation is partial but still very significant. This problem of properly capturing pressure drag was raised previously in the near-field force analysis in Section 5.1.2.

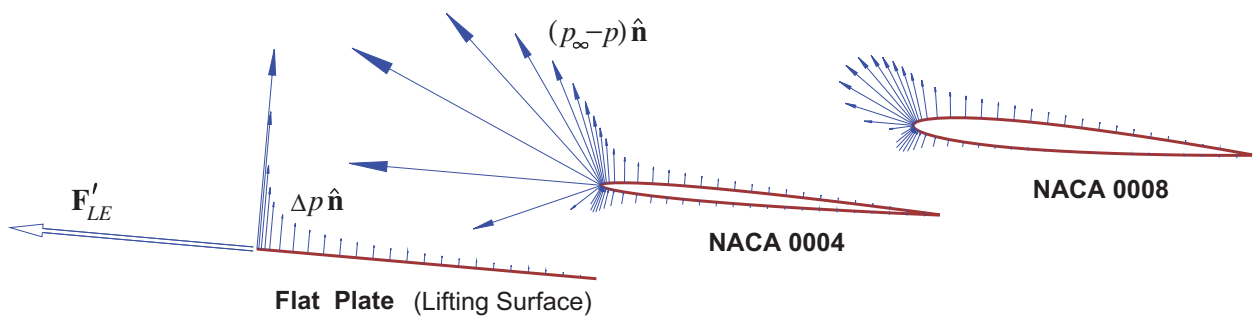

Figure 6.5: Surface pressure vectors on inviscid NACA 0008 and NACA 0004 airfoils at $\alpha = 5°$ computed with a panel method. In the zero thickness limit the leading edge suction becomes a tangential force $\mathbf{F}'_{s_{LE}}$, and the remaining pressure loads are represented by the $\Delta p\,\hat{\mathbf{n}}$ distribution.

The leading edge suction force can be determined by applying the integral momentum theorem to a control volume enclosing the leading edge point, and assuming that the local vortex sheet strength varies as

$$\gamma(x) \;\simeq\; \frac{C}{\sqrt{x}}$$

where x is the distance from the leading edge and C is some constant. The resulting suction force per leading edge arc length is

$$\begin{aligned} \mathbf{F}'_{s_{LE}} &= \frac{\pi}{4}\,\rho C^2\,\hat{\mathbf{t}} \\ C &= \lim_{x\to 0}\left(\gamma\,\sqrt{x}\,\right) \end{aligned} \tag{6.23}$$

where C is now determined from the actual sheet strength $\gamma(s,\ell)$ of the solution, and $\hat{\mathbf{t}}$ is the unit tangential vector pointing ahead of the leading edge, and normal to both $\hat{\mathbf{n}}$ and the leading edge line.

The overall force and moment on the configuration from the normal forces are finally obtained by integration of the pressure loading distribution and also the leading edge suction forces over all the surfaces.

$$\mathbf{F} \;=\; \sum_{\text{surfaces}} \iint_{\text{surface}} \Delta p\;\hat{\mathbf{n}}\;\mathrm{d}s\,\mathrm{d}\ell \;+\; \int_{\text{L.E.}} \mathbf{F}'_{s_{LE}}\,\mathrm{d}s \tag{6.24}$$

$$\mathbf{M} \;=\; \sum_{\text{surfaces}} \iint_{\text{surface}} \Delta p\;\mathbf{r}\times\hat{\mathbf{n}}\;\mathrm{d}s\,\mathrm{d}\ell \;+\; \int_{\text{L.E.}} \mathbf{r}\times\mathbf{F}'_{s_{LE}}\,\mathrm{d}s \tag{6.25}$$

The integration over the wakes is not performed. The rationale is that a real wake must have $\Delta p = 0$ anyway, even if this isn't quite true in the simplified lifting-surface model in which the wake γ is assumed to be aligned with $\hat{\mathbf{x}}$. To get a truly force-free wake in the model would require aligning the wake geometry and its strength γ with the local $\mathbf{V}_a$ direction. This would make the lifting surface problem nonlinear, since the geometry of the vortex sheets would then depend on the flow solution itself. All these complications are sidestepped by assuming the fixed $\hat{\mathbf{x}}$ wake direction and simply ignoring the resulting implied wake loads.

6.4.4 Trefftz-plane loads

An alternative to the above near-field force calculation is to use the far-field or Trefftz-plane approach. In general, this is a more reliable approach especially for the D_i component, since it avoids the usual pressure-drag cancellation errors discussed in Chapter 5, these being especially problematic with the simple leading edge suction model.

Referring to Figure 5.7, the first step here is to compute the Trefftz-plane perturbation velocity

$$\nabla\varphi(s) \;=\; \sum_{\text{sheets}} \frac{1}{2\pi} \int_0^{s_{\max}} \gamma(s') \, \frac{\hat{\mathbf{x}} \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^2} \, \mathrm{d}s' \tag{6.26}$$

at each sheet location $\mathbf{r}(s)$, where the integral is taken along all sheet traces shown in Figure 6.4. The wake sheet potential jump is also required.

$$\Delta\varphi(s) \;=\; \int_0^s \gamma(s') \; \mathrm{d}s' \tag{6.27}$$

$$\text{or} \qquad \Delta\varphi(s) \;=\; \mu(s) \tag{6.28}$$

The induced drag is then computed directly via expression (5.47).

$$D_i \;=\; \sum_{\text{sheets}} -\frac{1}{2}\rho \int_0^{s_{\max}} \Delta\varphi \;\; \nabla\varphi \cdot \hat{\mathbf{n}} \; \mathrm{d}s \tag{6.29}$$

The Trefftz plane also provides alternative means of computing the total lift and sideforce via expressions (5.64) and (5.65).

$$Y \;=\; \sum_{\text{sheets}} \rho V_\infty \int_0^{s_{\max}} -\Delta\varphi \;\; n_y \; \mathrm{d}s \tag{6.30}$$

$$L \;=\; \sum_{\text{sheets}} \rho V_\infty \int_0^{s_{\max}} \Delta\varphi \;\; n_z \; \mathrm{d}s \tag{6.31}$$

Note however, that this Trefftz plane force calculation method gives only the <u>total</u> forces. Equations (6.24) and (6.25) must be used for moments, and also for forces on the individual surfaces.

6.5 Vortex Lattice Method

The Vortex Lattice (VL) method is a numerical solution implementation of the general 3D lifting surface problem described above. It is also the simplest general 3D potential flow calculation method. It is commonly used in initial aircraft configuration development, where its simplicity and speed allow a large number of configurations to be examined. It is also used for initial structural load estimation, and to provide the trim state values and stability and control derivatives for the linearized force and moment equations (6.12), (6.13).

6.5.1 Vortex lattice discretization

The VL method discretizes the vortex-sheet strength distribution on each lifting surface and its wake by lumping it into a collection of *horseshoe vortices*, as shown in Figure 6.6. Each horseshoe vortex (h.v.) consists of three straight legs, or segments: a *bound leg* which lies on the surface, and two *trailing legs* extending from the bound leg's endpoints to downstream infinity and parallel to the x axis. All three legs of the i'th h.v. have the same constant circulation strength Γ_i.

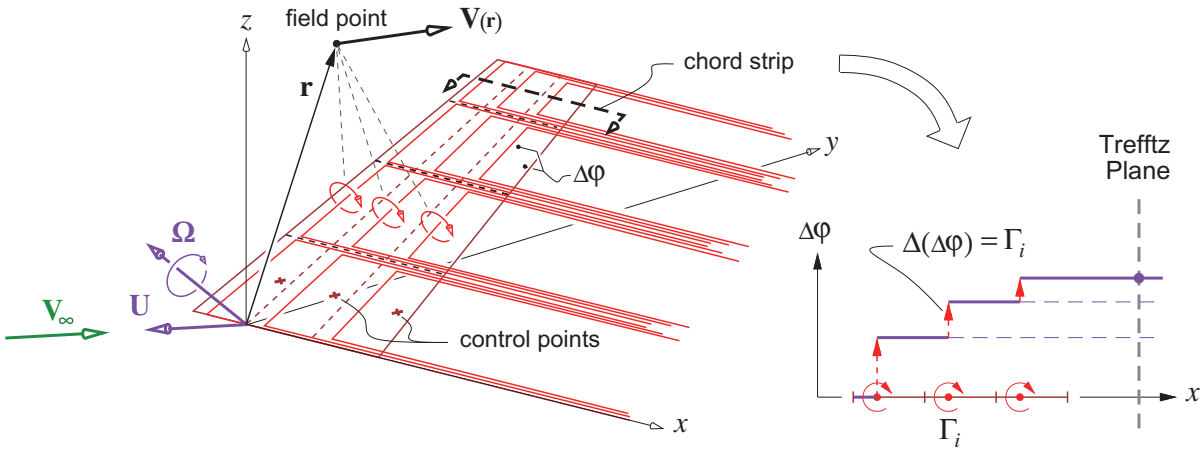

Figure 6.6: 3D lifting surface discretized by a Vortex Lattice of horseshoe vortices, which all contribute to the velocity $\mathbf{V}$ at any field point $\mathbf{r}$. Setting $\mathbf{r}$ at each control point allows imposition of flow tangency there. Each horseshoe vortex i also adds a contribution of Γ_i to the total potential jump $\Delta\varphi$ within its perimeter. Total accumulated $\Delta\varphi$ along a chord strip is the Trefftz-plane value shown in Figure 5.12.

In a discrete sense, this configuration of vortices satisfies the zero-divergence requirement on the vortex sheet strength $\boldsymbol{\gamma}(s,\ell)$ which is discussed in Section 2.4, since any circuit drawn on the surface will have a filament with a fixed circulation both entering and leaving it. Note also that each h.v. adds zero net circulation in the Trefftz plane, where its two trailing legs have equal and opposite circulations.

An equivalent interpretation of the h.v. configuration is a piecewise-constant potential jump or normal-doublet distribution. Each h.v. contributes $\Delta(\Delta\varphi) = \Gamma_i$ to the total potential jump $\Delta\varphi$ within its perimeter. The total circulation of all the h.v.'s in a chord strip then gives the $\Delta\varphi$ in the Trefftz plane, as shown in Figure 6.6. This $\Delta\varphi$ is also shown in Figure 5.12, where it's used to construct the Trefftz-plane velocity and evaluate the Trefftz-plane forces.

6.5.2 Velocity field representation

The overall velocity field relative to the VL configuration at any point $\mathbf{r}$ is given by relation (6.18), where the $\boldsymbol{\gamma}(s,\ell)$ vortex-sheet strength is now lumped into the collection of h.v. filaments, with each filament having a constant strength Γ_i. The surface integral in (6.17) is therefore replaced with a collection of Biot-Savart line integrals, one for each h.v.

$$\begin{aligned}\mathbf{V}(\mathbf{r}) &= \sum_{\text{h.v.'s}} \frac{\Gamma}{4\pi} \int \frac{\mathrm{d}\boldsymbol{\ell}' \times (\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|^3} - (\mathbf{U} + \boldsymbol{\Omega}\times\mathbf{r}) \\ &= \sum_{i=1}^{N} \Gamma_i \, \hat{\mathbf{V}}_i(\mathbf{r}) - (\mathbf{U} + \boldsymbol{\Omega}\times\mathbf{r}) \qquad (6.32)\end{aligned}$$

Evaluation of the Biot-Savart integral of the i'th h.v. has produced the $\hat{\mathbf{V}}_i$ kernel function, given by the following expression. The $\mathbf{a}$ and $\mathbf{b}$ vectors are shown in Figure 6.7.

$$\hat{\mathbf{V}}_{i(\mathbf{r})} = \frac{1}{4\pi}\left\{ \frac{\mathbf{a}\times\mathbf{b}}{|\mathbf{a}||\mathbf{b}|+\mathbf{a}\cdot\mathbf{b}}\left(\frac{1}{|\mathbf{a}|}+\frac{1}{|\mathbf{b}|}\right) + \frac{\mathbf{a}\times\hat{\mathbf{x}}}{|\mathbf{a}|-\mathbf{a}\cdot\hat{\mathbf{x}}}\frac{1}{|\mathbf{a}|} - \frac{\mathbf{b}\times\hat{\mathbf{x}}}{|\mathbf{b}|-\mathbf{b}\cdot\hat{\mathbf{x}}}\frac{1}{|\mathbf{b}|} \right\} \tag{6.33}$$

The three terms in (6.33) correspond to the bound leg, the $\mathbf{r}_{\mathrm{a}}$-point trailing leg, and the $\mathbf{r}_{\mathrm{b}}$-point trailing leg, respectively. Note that $\hat{\mathbf{V}}_i$ has units of $1/\text{length}$.

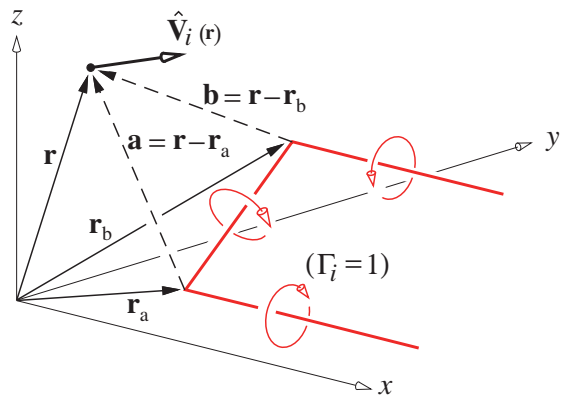

Figure 6.7: Geometry of one horseshoe vortex, producing a unit-strength velocity $\hat{\mathbf{V}}_{i(\mathbf{r})}$ at some field point $\mathbf{r}$.

6.5.3 Flow tangency condition

The flow tangency condition (6.19) is imposed at the N control points by choosing $\mathbf{r}$ in (6.32) to be the control point $\mathbf{r}_i^c$ of each h.v. in turn, and setting the resulting normal velocity component to zero.

$$\mathbf{V}_{(\mathbf{r}_i^c)}\cdot\mathbf{n}_{i(\delta_l)} = \left(\sum_{j=1}^{N}\Gamma_j\,\hat{\mathbf{V}}_{j(\mathbf{r}_i^c)} - (\mathbf{U}+\mathbf{\Omega}\times\mathbf{r}_i^c)\right)\cdot\mathbf{n}_{i(\delta_l)} = 0 \qquad (i=1\ldots N) \tag{6.34}$$

To avoid conflict with the control point index i, the summation index over the h.v.'s has been changed to j.

The normal vector $\mathbf{n}_i$ depends on δ_l, which are control variables which define the deflections of some number of control surfaces. The control index $l = 1, 2\ldots N_l$ is more practical for computation than the earlier $\delta_a, \delta_e\ldots$ notation introduced in Section 6.3.2.

The deflections are modeled by rotating each $\mathbf{n}_i$ on that control surface about a specified hinge axis. In keeping with the small-angle approximations used throughout lifting surface theory and the VL method in particular, each normal vector's dependence on δ_l is linearized. Referring to Figure 6.8 we have

$$\mathbf{n}_{i(\delta_l)} \simeq \mathbf{n}_{0_i} + \sum_{l=1}^{N_l}\mathbf{n}_{l_i}\,\delta_l \tag{6.35}$$

$$\mathbf{n}_{l_i} \equiv \frac{\partial\mathbf{n}_i}{\partial\delta_l} = g_l\,\hat{\mathbf{h}}_{l_i}\times\mathbf{n}_{0_i} \tag{6.36}$$

where $\hat{\mathbf{h}}_l$ is the hinge-axis unit vector about which $\mathbf{n}$ rotates in response to the δ_l control deflection, and g_l is the "control gain," included so that the δ_l control variable doesn't have to be the actual local deflection angle in radians. To linearize the control influence we will assume that $-U_x\hat{\mathbf{x}} \simeq V_\infty\,\hat{\mathbf{x}}$ which is equivalent to

$\alpha, \beta \ll 1$, and that the normal-vector control deflections $\mathbf{n}_{l_i}\delta_l$ are small compared to the undeflected normal vector $\mathbf{n}_{0_i}$. Thus the flow-tangency equations (6.34) are approximated by

$$\left(\sum_{j=1}^{N} \Gamma_j \, \hat{\mathbf{V}}_{j(\mathbf{r}_i^c)} \; - \; (\mathbf{U} + \mathbf{\Omega}\times\mathbf{r}_i^c)\right) \cdot \mathbf{n}_{0_i} \; + \; \sum_{l=1}^{N_l} V_\infty \, \hat{\mathbf{x}} \cdot \mathbf{n}_{l_i} \, \delta_l \; = \; 0 \qquad (i = 1 \ldots N) \quad (6.37)$$

where all products of two small quantities have been dropped.

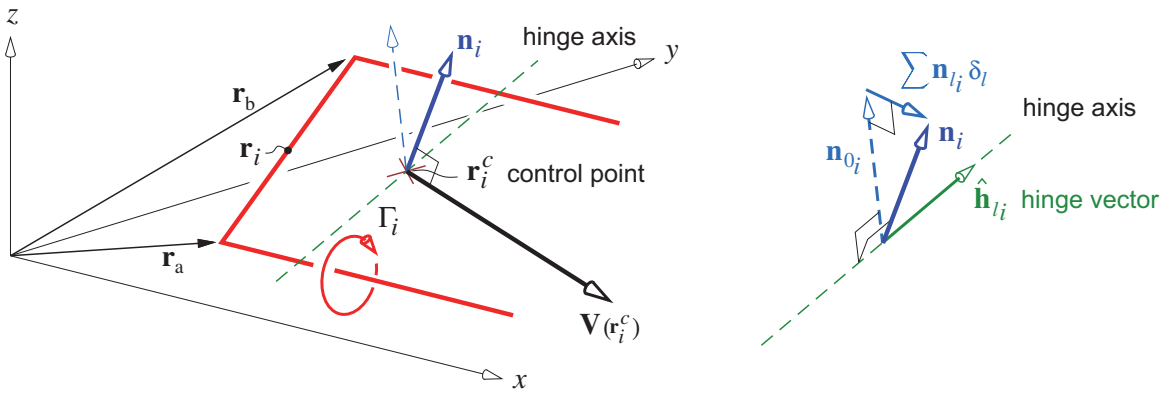

Figure 6.8: Flow tangency imposed at control point of i'th horseshoe vortex, biased by rotation of the normal vector $\mathbf{n}_i$ via the control variables δ_l.

6.5.4 Linear system setup and solution

The subsequent numerical implementation of equations (6.37) will be done in terms of the following variables normalized with V_∞, denoted by the overbar $\bar{()}$. Note that $\bar{\mathbf{U}}$ is then dimensionless, while $\bar{\Gamma}$ and $\bar{\mathbf{\Omega}}$ have units of length and $1/$length, respectively.

$$\bar{\Gamma}_i \equiv \frac{\Gamma_i}{V_\infty} \quad , \quad \bar{\mathbf{U}} \equiv \frac{\mathbf{U}}{V_\infty} = \begin{Bmatrix} \bar{U}_x \\ \bar{U}_y \\ \bar{U}_z \end{Bmatrix} = \begin{Bmatrix} -\cos\alpha\cos\beta \\ \sin\beta \\ -\sin\alpha\cos\beta \end{Bmatrix} \quad , \quad \bar{\mathbf{\Omega}} \equiv \frac{\mathbf{\Omega}}{V_\infty} = \begin{Bmatrix} \bar{\Omega}_x \\ \bar{\Omega}_y \\ \bar{\Omega}_z \end{Bmatrix} \qquad (6.38)$$

The linearized flow tangency conditions (6.37) constitute an $N \times N$ linear system for the $\bar{\Gamma}_i$ normalized vortex strengths when the $\bar{\mathbf{U}}, \bar{\mathbf{\Omega}}, \delta_l$ terms are placed on the righthand side.

$$\begin{aligned} \Big[\; A_{ij} \;\Big]\Big\{\bar{\Gamma}_j\Big\} \; &= \; \bar{U}_x\Big\{\; \hat{\mathbf{x}} \cdot \mathbf{n}_{0_i} \;\Big\} \; + \; \bar{U}_y\Big\{\; \hat{\mathbf{y}} \cdot \mathbf{n}_{0_i} \;\Big\} \; + \; \bar{U}_z\Big\{\; \hat{\mathbf{z}} \cdot \mathbf{n}_{0_i} \;\Big\} \\ &+ \; \bar{\Omega}_x\Big\{\hat{\mathbf{x}}\times\mathbf{r}_i^c \cdot \mathbf{n}_{0_i}\Big\} + \bar{\Omega}_y\Big\{\hat{\mathbf{y}}\times\mathbf{r}_i^c \cdot \mathbf{n}_{0_i}\Big\} + \bar{\Omega}_z\Big\{\hat{\mathbf{z}}\times\mathbf{r}_i^c \cdot \mathbf{n}_{0_i}\Big\} \\ &+ \; \delta_1\Big\{-\hat{\mathbf{x}} \cdot \mathbf{n}_{1_i}\Big\} \; + \; \delta_2\Big\{-\hat{\mathbf{x}} \cdot \mathbf{n}_{2_i}\Big\} \; + \; \ldots \; + \; \delta_{N_l}\Big\{-\hat{\mathbf{x}} \cdot \mathbf{n}_{Nl_i}\Big\} \qquad (6.39) \\ A_{ij} \; &\equiv \; \hat{\mathbf{V}}_{j(\mathbf{r}_i^c)} \cdot \mathbf{n}_{0_i} \qquad (6.40) \end{aligned}$$

The Aerodynamic Influence Coefficient matrix A_{ij} and the righthand side vectors in braces are functions of the vortex lattice geometry only, and hence are known a priori.

Multiplying (6.39) through by A_{ij}^{-1} using LU-factorization and back-substitution gives the solution vector $\bar{\Gamma}_i$ as a sum of known $6+N_l$ independent vectors, whose coefficients (arbitrary at this point) are the operating

parameters $\bar{U}_x, \bar{U}_y \ldots \delta_{N_l}$. When these parameters are specified, $\bar{\Gamma}_i$ is determined by summing all $6+N_l$ vectors.

$$\left\{\bar{\Gamma}_i\right\} = \bar{U}_x\left\{\bar{\Gamma}_{1_i}\right\} + \bar{U}_y\left\{\bar{\Gamma}_{2_i}\right\} + \bar{U}_z\left\{\bar{\Gamma}_{3_i}\right\} + \bar{\Omega}_x\left\{\bar{\Gamma}_{4_i}\right\} + \bar{\Omega}_y\left\{\bar{\Gamma}_{5_i}\right\} + \bar{\Omega}_z\left\{\bar{\Gamma}_{6_i}\right\} + \sum_{l=1}^{N_l}\delta_l\left\{\bar{\Gamma}_{6+l_i}\right\} \quad (6.41)$$

6.5.5 Near-field force and moment calculation

Once the $\bar{\Gamma}_i$ strengths are known, the normalized velocity relative to the midpoint $\mathbf{r}_i$ of the i'th h.v. is calculated, as shown in Figure 6.9. This uses the same form as (6.32),

$$\frac{\mathbf{V}_i}{V_\infty} \equiv \bar{\mathbf{V}}_i = \begin{Bmatrix}\bar{V}_{xi}\\ \bar{V}_{yi}\\ \bar{V}_{zi}\end{Bmatrix} = \sum_{j=1}^{N}\bar{\Gamma}_j\,\hat{\mathbf{V}}_{j(\mathbf{r}_i)} - \begin{Bmatrix}\bar{U}_x\\ \bar{U}_y\\ \bar{U}_z\end{Bmatrix} - \begin{Bmatrix}\bar{\Omega}_x\\ \bar{\Omega}_y\\ \bar{\Omega}_z\end{Bmatrix}\times\mathbf{r}_i \quad (6.42)$$

$$\mathbf{r}_i = \tfrac{1}{2}\left(\mathbf{r}_\mathrm{a}+\mathbf{r}_\mathrm{b}\right) \quad (6.43)$$

except the h.v. kernel functions $\hat{\mathbf{V}}_j$ here are different than those in the flow-tangency condition (6.34), since the bound-leg midpoint locations $\mathbf{r}_i$ are different from the control-point locations $\mathbf{r}_i^c$. In the $\hat{\mathbf{V}}_j$ function in (6.42) it is also necessary to omit the bound leg's contribution on itself, which is the first term in (6.33), since this is singular at $\mathbf{r}_i$.

The normalized force $\bar{\mathbf{F}}_i$ on each h.v. is computed by the integrated form of the local pressure loading relation (6.22).

$$\mathbf{F}_i = \iint_{\text{element}}\Delta p\,\hat{\mathbf{n}}\;\mathrm{d}\mathcal{S} = \iint_{\text{element}}\rho\,\mathbf{V}_a\times\boldsymbol{\gamma}\;\mathrm{d}\mathcal{S} = \rho\mathbf{V}_i\times\boldsymbol{\ell}_i\,\Gamma_i \quad (6.44)$$

$$\text{where}\qquad \boldsymbol{\ell}_i = \mathbf{r}_\mathrm{b}-\mathbf{r}_\mathrm{a}$$

$$\bar{\mathbf{F}}_i \equiv \frac{\mathbf{F}_i}{\frac{1}{2}\rho_\infty V_\infty^2 S_\mathrm{ref}} \simeq \frac{2}{S_\mathrm{ref}}\bar{\mathbf{V}}_i\times\boldsymbol{\ell}_i\,\bar{\Gamma}_i \quad (6.45)$$

The last step in (6.44) consists of lumping the vortex sheet on the element into the element h.v.'s bound leg vortex segment, $\iint\boldsymbol{\gamma}\,\mathrm{d}\mathcal{S}\simeq\boldsymbol{\ell}_i\Gamma_i$. The relevant quantities are shown in Figure 6.9.

Interestingly enough, the Kutta-Joukowsky force calculation form (6.45) gives exactly zero drag in the 2D case where there are no trailing h.v. legs. Therefore it implicitly accounts for the leading edge suction force, which then does not need to be added explicitly.

The total normalized force and moment on the whole configuration are obtained by summation of all the individual h.v. contributions. The moment is defined about a specified point $\mathbf{r}_\mathrm{ref}$.

$$\bar{\mathbf{F}} = \sum_{i=1}^{N}\bar{\mathbf{F}}_i \quad (6.46)$$

$$\bar{\mathbf{M}} = \sum_{i=1}^{N}(\mathbf{r}_i-\mathbf{r}_\mathrm{ref})\times\bar{\mathbf{F}}_i \quad (6.47)$$

The standard dimensionless force and moment coefficients in stability axes, shown in Figure 6.2, are obtained by rotating $\bar{\mathbf{F}}$ and $\bar{\mathbf{M}}$ using the $\bar{\bar{\mathbf{T}}}^\mathrm{s}$ matrix given by (6.5). The reference span and chord $b_\mathrm{ref}, c_\mathrm{ref}$ are

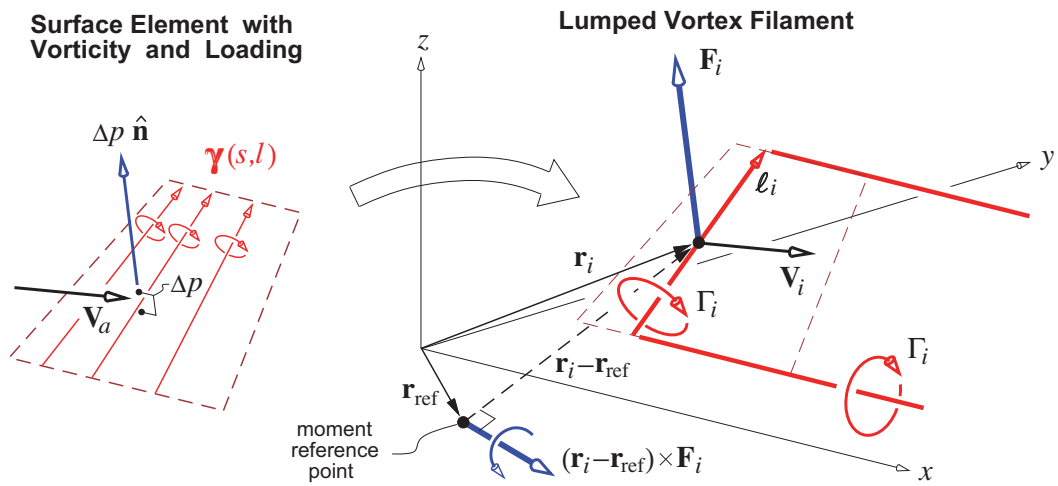

Figure 6.9: Net pressure loading on a surface element approximated by the Kutta-Joukowsky force on the element's horseshoe vortex bound leg $\boldsymbol{\ell}_i$.

also used here to non-dimensionalize the moments.

$$\begin{Bmatrix} C_{D_i} \\ C_Y \\ C_L \end{Bmatrix} = \begin{bmatrix} & \\ & \bar{\bar{\mathbf{T}}}^{\mathrm{s}} & \\ & \end{bmatrix} \begin{Bmatrix} \bar{F}_x \\ \bar{F}_y \\ \bar{F}_z \end{Bmatrix} \tag{6.48}$$

$$\begin{Bmatrix} C'_\ell \\ C'_m \\ C'_n \end{Bmatrix} = \begin{bmatrix} & \\ & \bar{\bar{\mathbf{T}}}^{\mathrm{s}} & \\ & \end{bmatrix} \begin{Bmatrix} -\bar{M}_x/b_{\text{ref}} \\ \bar{M}_y/c_{\text{ref}} \\ -\bar{M}_z/b_{\text{ref}} \end{Bmatrix} \tag{6.49}$$

If there is a nonzero sideslip, $\beta \neq 0$, then the physically correct expression for the induced drag coefficient is given by the wind-axis relation (6.8).

$$C_{D_i} = -\bar{\mathbf{F}} \cdot \bar{\mathbf{U}} \tag{6.50}$$

6.5.6 Trefftz-plane force calculation

As discussed in Section 6.4.4, the Trefftz-plane provides an alternative to the near-field result (6.6) for calculating the wind-axes forces. A suitable discrete formulation of the Trefftz-plane integrals which is applicable to the VL method here was already given in Section 5.8.2. To apply those results it is first necessary to define the piecewise-constant wake potential jump distribution $\Delta\varphi_i$ for each h.v. "chord strip," shown in Figure 6.6, whose h.v.'s have their trailing legs superimposed in the Trefftz-plane.

$$\Delta\bar{\varphi}_i = \sum_{\text{strip}} \bar{\Gamma}_i \tag{6.51}$$

The summation is only over the h.v.'s within that chord strip, indicated in Figure 6.6. The normalized version of expression (5.81) for the wake-normal velocity $\partial\bar{\varphi}/\partial n_i$ and expressions (5.77), (5.78), (5.82) for the forces Y, L, D_i can then be used as written.

6.5.7 Stability and control derivative calculation

The VL method is well suited to rapid calculation of stability and control derivatives in the small-angle operating range $\alpha, \beta, \bar{p}, \bar{q}, \bar{r} \ll 1$. The calculation can be performed by finite-differencing slightly perturbed flow solutions, e.g.

$$C_{L_\alpha} \simeq \frac{C_L(\alpha_0+\Delta\alpha\,,\beta_0) - C_L(\alpha_0\,,\beta_0)}{\Delta\alpha} \quad , \qquad C_{n_\beta} \simeq \frac{C_n(\alpha_0\,,\beta_0+\Delta\beta) - C_n(\alpha_0\,,\beta_0)}{\Delta\beta} \tag{6.52}$$

A more economical alternative is to implicitly differentiate the overall force and moment summations with respect to each parameter via the chain rule, noting that $\bar{\Gamma}_i$ as given by (6.41) and $\bar{\mathbf{V}}_i$ as given by (6.42) have relatively simple dependencies on the parameters. For example, the α-derivatives of the following quantities can be evaluated in parallel with each quantity itself.

$$\frac{\partial \bar{\mathbf{U}}}{\partial \alpha} = \begin{Bmatrix} \sin\alpha\cos\beta \\ 0 \\ -\cos\alpha\cos\beta \end{Bmatrix} \tag{6.53}$$

$$\frac{\partial \bar{\Gamma}_i}{\partial \alpha} = \frac{\partial \bar{U}_x}{\partial \alpha}\,\bar{\Gamma}_{1_i} + \frac{\partial \bar{U}_z}{\partial \alpha}\,\bar{\Gamma}_{3_i} \tag{6.54}$$

$$\frac{\partial \bar{\mathbf{V}}_i}{\partial \alpha} = \sum_{j=1}^{N} \frac{\partial \bar{\Gamma}_j}{\partial \alpha}\,\hat{\mathbf{V}}_{j(\mathbf{r}_i)} - \frac{\partial \bar{\mathbf{U}}}{\partial \alpha} \tag{6.55}$$

$$\frac{\partial \bar{\mathbf{F}}_i}{\partial \alpha} = \frac{2}{S_{\text{ref}}}\left(\frac{\partial \bar{\mathbf{V}}_i}{\partial \alpha} \times \boldsymbol{\ell}_i\,\bar{\Gamma}_i + \bar{\mathbf{V}}_i \times \boldsymbol{\ell}_i\,\frac{\partial \bar{\Gamma}_i}{\partial \alpha}\right) \tag{6.56}$$

$$\frac{\partial \bar{\mathbf{F}}}{\partial \alpha} = \sum_{i=1}^{N} \frac{\partial \bar{\mathbf{F}}_i}{\partial \alpha} \tag{6.57}$$

$$\frac{\partial}{\partial \alpha}\begin{Bmatrix} C_{D_i} \\ C_Y \\ C_L \end{Bmatrix} = \frac{\mathrm{d}\bar{\bar{\mathbf{T}}}^{\mathrm{s}}}{\mathrm{d}\alpha}\,\bar{\mathbf{F}} + \bar{\bar{\mathbf{T}}}^{\mathrm{s}}\,\frac{\partial \bar{\mathbf{F}}}{\partial \alpha} \tag{6.58}$$

Note how each derivative calculation uses the derivatives calculated earlier. The final result (6.58) is the sought-after force stability derivatives with respect to α. The same procedure is used for the moment derivative vector. The procedure is also repeated for all the remaining parameters $\beta, \bar{p}, \bar{q}, \bar{r}, \delta_l$. The advantage of this direct differentiation method over the finite-difference approach (6.52) is that it is economical and effectively exact.

6.6 Slender Body Theory

Slender Body Theory is applicable to bodies such as slender fuselages and nacelles, and also to very slender delta wings with $AR \ll 1$. The key simplifying assumption, and the definition of "slender," is that the yz-plane cross sections of the body and the flow vary slowly in the streamwise x direction relative to the y, z directions. This implies that the streamwise component φ_x of the perturbation velocity $\nabla\varphi$ is negligible compared to the transverse components φ_y, φ_z.

$$\varphi_x \ll \varphi_y\,,\,\varphi_z \tag{6.59}$$

It's useful to note the similarities with the Trefftz plane, introduced in Section 5.6.

6.6.1 Slender body geometry

The geometry of the body is specified by its surface position vector

$$\tilde{\mathbf{r}}_{(x,s)} = x\,\hat{\mathbf{x}} + \left[\,\tilde{y}_{(s\,;x)}\,\right]\hat{\mathbf{y}} + \left[\,\tilde{z}_{(s\,;x)}\,\right]\hat{\mathbf{z}} \tag{6.60}$$

where s is some parameter, such as the arc length in the yz plane. This defines the $\tilde{y}, \tilde{z}$ cross-section shape of the body at some x location, as shown in Figure 6.10. Since the vectors $\partial\tilde{\mathbf{r}}/\partial s$ and $\partial\tilde{\mathbf{r}}/\partial x$ are both tangent to the body surface, the unit normal vector can be determined via their cross product.

$$\mathbf{n}_{(x,s)} \;=\; \frac{\partial\tilde{\mathbf{r}}}{\partial s}\times\frac{\partial\tilde{\mathbf{r}}}{\partial x} \;=\; \left[\frac{\partial\tilde{y}}{\partial s}\frac{\partial\tilde{z}}{\partial x} - \frac{\partial\tilde{z}}{\partial s}\frac{\partial\tilde{y}}{\partial x}\right]\hat{\mathbf{x}} \;+\; \frac{\partial\tilde{z}}{\partial s}\,\hat{\mathbf{y}} \;-\; \frac{\partial\tilde{y}}{\partial s}\,\hat{\mathbf{z}} \qquad , \qquad \hat{\mathbf{n}} \;=\; \frac{\mathbf{n}}{|\mathbf{n}|} \tag{6.61}$$

In practice, the dimensional normal vector $\mathbf{n}$ is simpler and is adequate for the construction of flow tangency boundary conditions.

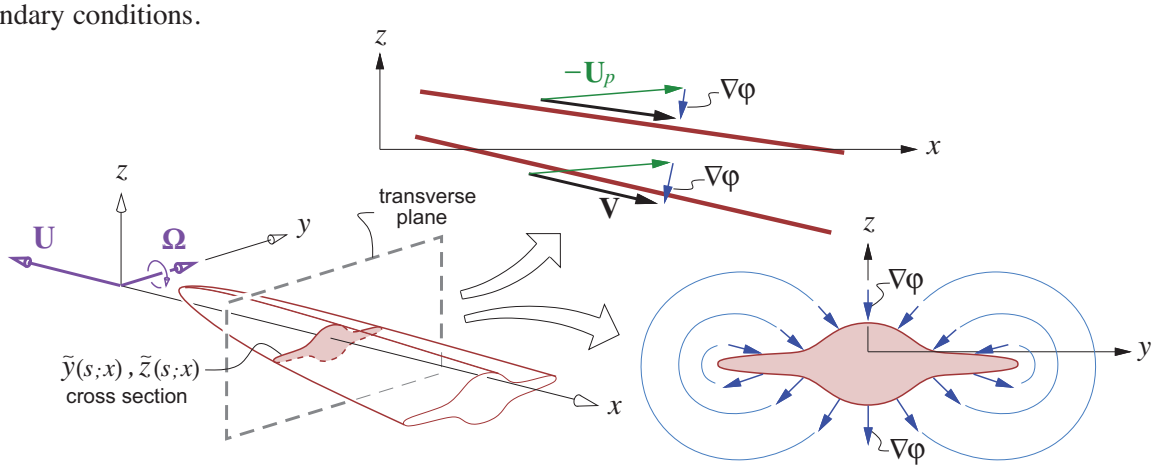

Figure 6.10: Slender body's perturbation velocity $\nabla\varphi$ is nearly two-dimensional in the transverse yz plane. Body cross section at each station x is given by the $\tilde{y}_{(s\,;\,x)}$, $\tilde{z}_{(s\,;\,x)}$ coordinate functions.

6.6.2 Slender body flow-field

The body is assumed to move at steady velocity $\mathbf{U} = -V_\infty\,\hat{\mathbf{x}}$ and pitch rate $\mathbf{\Omega} = q\,\hat{\mathbf{y}}$, so the local apparent freestream is $-\mathbf{U}_p = -(\mathbf{U} + \mathbf{\Omega}\times\mathbf{r}) = V_\infty\,\hat{\mathbf{x}} + qx\,\hat{\mathbf{z}}$. Any angle of attack or sideslip is assumed to be included in the $\tilde{\mathbf{r}}$ geometry definition. The perturbation potential $\varphi_{(x,y,z)}$ of the flow about the slender body is assumed to locally satisfy the <u>two-dimensional</u> crossflow Laplace's equation

$$\varphi_{yy} \;+\; \varphi_{zz} \;=\; 0 \qquad\qquad (\text{ for } y^2+z^2 \ll \ell^2\,) \tag{6.62}$$

where ℓ is the length of the body. Dropping the φ_{xx} term has been justified by the slender-flow assumption (6.59). The appropriate boundary conditions are flow tangency of the total velocity on the body surface, and asymptotically zero perturbation flow far away.

$$[\,(V_\infty+\varphi_x)\,\hat{\mathbf{x}} + \varphi_y\,\hat{\mathbf{y}} + (qx+\varphi_z)\,\hat{\mathbf{z}}\,]\cdot\mathbf{n} \;=\; 0$$

$$\text{or} \qquad \varphi_y\,n_y \;+\; \varphi_z\,n_z \;\simeq\; -(V_\infty\,n_x + qx\,n_z) \qquad (\text{ on body }) \tag{6.63}$$

$$\text{and} \qquad \varphi \;\to\; 0 \qquad (\text{for } y^2+z^2 \gg \ell^2) \tag{6.64}$$

The $\varphi_x n_x$ term has been dropped from the right side of (6.63) since it is negligible relative to the other terms.

Laplace's equation (6.62) and boundary conditions (6.63),(6.64) define a 2D incompressible potential flow with a known nonzero normal velocity $-(V_\infty\,n_x + qx\,n_z)$ which is due to the pitch rate and also the inclination of the body surface relative to the freestream x direction. Such a flow is sketched in Figure 6.10 on bottom right. It can be computed for a given arbitrary cross-section using any standard 2D panel method, but with a different righthand side than usual. To compute the overall 3D flow, some sufficiently large number of individual 2D problems would actually have to be solved, one for each discrete x location.

Once the potential $\varphi(x,y,z)$ is computed, the pressure is determined using the Bernoulli equation.

$$\begin{aligned} p \,+\, \tfrac{1}{2}\rho\left[(V_\infty+\varphi_x)^2 \,+\, \varphi_y^2 \,+\, \varphi_z^2\right] &= p_\infty \,+\, \tfrac{1}{2}\rho V_\infty^2 \\ p - p_\infty &= -\rho V_\infty \varphi_x \,-\, \tfrac{1}{2}\rho\left(\varphi_y^2 \,+\, \varphi_z^2\right) \end{aligned} \tag{6.65}$$

Even though the streamwise perturbation velocity φ_x was neglected in the local 2D problem defined by Laplace's equation (6.62) and boundary condition (6.63), it is required for the pressure calculation. The reason is that $V_\infty\varphi_x$ is not necessarily negligible compared to $\varphi_y^2+\varphi_z^2$ and hence must be retained here.

6.6.3 2D unsteady flow interpretation

The sequence of solutions along x can be interpreted as a 2D unsteady flow $\varphi(t,y,z)$ if we make the Galilean transformation

$$x \,=\, V_\infty\, t \quad , \qquad \frac{\partial}{\partial x} \,=\, \frac{1}{V_\infty}\frac{\partial}{\partial t}$$

where the new observer is traveling along the x axis at the freestream speed V_∞. The $\tilde{y}(s\,;t)$, $\tilde{z}(s\,;t)$ cross-sectional shape in the transverse plane and the resulting transverse flow then appear to change in time. In this unsteady interpretation the streamwise perturbation velocity transforms as $\varphi_x \to \varphi_t/V_\infty$, which converts the steady 3D Bernoulli equation (6.65) into its 2D unsteady form which gives the same pressure.

$$p - p_\infty \;=\; -\rho\,\varphi_t \,-\, \tfrac{1}{2}\rho\left(\varphi_y^2 \,+\, \varphi_z^2\right)$$

In practice, when solving the 2D Laplace problem (6.62)–(6.64), the spatial sequence in x or the time sequence in t are computationally equivalent. And for computing the pressure, the 3D steady or the 2D unsteady Bernoulli equations are equivalent as well. Therefore, this transformation and unsteady interpretation does not provide any computational advantages, but it does give some additional insight into the problem.

6.6.4 Local 2D far-field

At intermediate distances which are large compared to the local body y, z dimension but small compared to the body length ℓ, the perturbation potential must have a local 2D far-field form as given by (2.78).

$$\varphi(r,\theta\,;x) \;=\; \frac{\Lambda(x)}{2\pi}\,\ln r \;+\; \frac{\kappa_z(x)}{2\pi}\,\frac{\sin\theta}{r} \tag{6.66}$$

$$\text{where} \qquad \begin{aligned} r(x,y,z) &\equiv \sqrt{y^2+(z-z_0(x))^2} \\ \theta(x,y,z) &\equiv \arctan\frac{z-z_0(x)}{y} \end{aligned}$$

The vortex term was omitted since there cannot be an overall circulation about this 3D body. The y doublet was omitted by the assumption of left/right symmetry about $y=0$, and it can always be eliminated in any case by rotating the axes so that the z axis aligns with the doublet-vector axis. Also, for generality the singularities are placed at some location $z=z_0(x)$ rather than at $z=0$.

The source strength is related to the body's cross-sectional area $A(x)$ as derived in Section 2.12.

$$\Lambda(x) \;=\; V_\infty\,\frac{\mathrm{d}A}{\mathrm{d}x} \tag{6.67}$$

The z-doublet is related to the lift via the the far-field lift integral (5.50). This requires the pressure, which is obtained from the Bernoulli equation (6.65),

$$p_\infty - p \;=\; \rho V_\infty\,\varphi_x \;=\; \rho V_\infty\,\frac{\mathrm{d}\kappa_z}{\mathrm{d}x}\,\frac{\sin\theta}{2\pi r} \tag{6.68}$$

The quadratic terms in (6.65) have been omitted here, since they become negligible at a sufficiently large control volume. Following the procedure in Section C.4, integral (5.50) is now evaluated on a $\mathrm{d}x$–long circular control volume of some radius r, whose arc length element is $\mathrm{d}l = r\,\mathrm{d}\theta$.

$$\begin{aligned}\frac{\mathrm{d}L}{\mathrm{d}x} &= \oint \left[\,(p_\infty - p)\,n_z \;-\; \rho\,\varphi_z\,\varphi_r\,\right]\,\mathrm{d}l \;=\; \rho V_\infty \frac{\mathrm{d}\kappa_z}{\mathrm{d}x}\int_0^{2\pi}\frac{\sin^2\theta}{2\pi r}\,r\,\mathrm{d}\theta \\ &= \frac{1}{2}\rho V_\infty \frac{\mathrm{d}\kappa_z}{\mathrm{d}x} \end{aligned} \tag{6.69}$$

Using (6.69) to calculate the overall lift of the body still requires relating κ_z to the body geometry, which in general is case-dependent. A simple geometry is considered next as an example.

6.6.5 Cambered body of revolution

A cambered body with circular cross sections is shown in Figure 6.11. Its surface geometry is defined by

$$\tilde{\mathbf{r}}_{(x,\theta)} \;=\; x\,\hat{\mathbf{x}} \;+\; \left[\,R\,\cos\theta\,\right]\,\hat{\mathbf{y}} \;+\; \left[\,R\,\sin\theta \;+\; Z - \alpha x\,\right]\,\hat{\mathbf{z}} \tag{6.70}$$

where $Z_{(x)}$ is the camber of the centerline, α is the overall body angle of attack, and $R_{(x)}$ is the radius of the local circular cross-section which is centered on $z = z_0{}_{(x)} = Z_{(x)} - \alpha x$.

The local normal vector is computed from (6.70) using the cross-product relation (6.61), with $\theta = s/R$.

$$\mathbf{n}_{(x,\theta)} \;=\; \left[-R\frac{\mathrm{d}R}{\mathrm{d}x} + \left(\alpha - \frac{\mathrm{d}Z}{\mathrm{d}x}\right) R\,\sin\theta\right]\,\hat{\mathbf{x}} \;+\; \Big[\,R\,\cos\theta\,\Big]\,\hat{\mathbf{y}} \;+\; \Big[\,R\,\sin\theta\,\Big]\,\hat{\mathbf{z}} \tag{6.71}$$

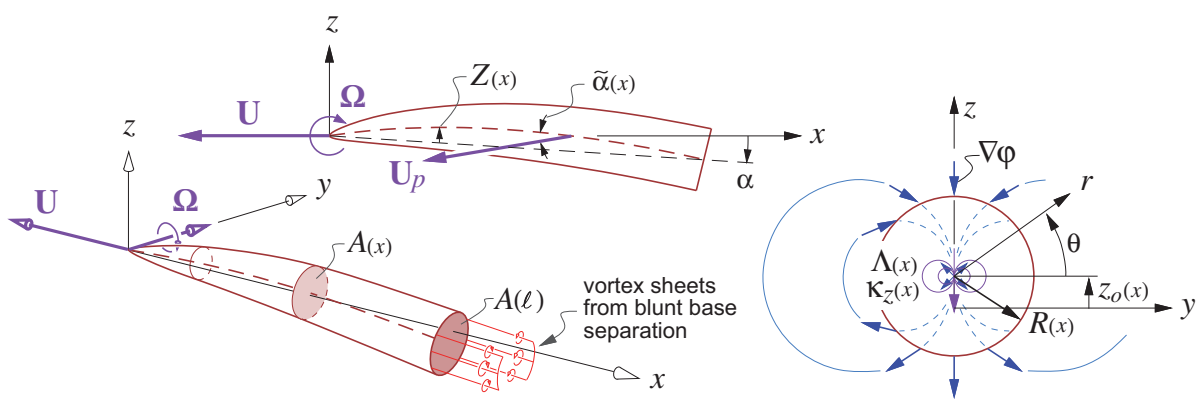

Figure 6.11: Slender body of revolution with overall angle of attack α, local added camber $Z_{(x)}$, and resulting local angle of attack $\tilde{\alpha}_{(x)}$. Local transverse flow is represented by source strength $\Lambda_{(x)}$ and z-doublet strength $\kappa_{z(x)}$, both located at the body center at $z_{0(x)}$. Vortex sheets are shed if a blunt base is present.

In terms of the cross-sectional area $A_{(x)} = \pi R^2$, the far-field source and doublet strengths are

$$\Lambda_{(x)} \;=\; 2\pi V_\infty R\frac{\mathrm{d}R}{\mathrm{d}x} \;=\; V_\infty\,\frac{\mathrm{d}A}{\mathrm{d}x} \tag{6.72}$$

$$\kappa_{z(x)} \;=\; 2\pi V_\infty\,R^2\,\tilde{\alpha} \;=\; 2V_\infty\,A\,\tilde{\alpha} \tag{6.73}$$

$$\tilde{\alpha}_{(x)} \;\equiv\; \alpha \;-\; \frac{\mathrm{d}Z}{\mathrm{d}x} + \frac{qx}{V_\infty} \tag{6.74}$$

where $\tilde{\alpha}$ is the <u>local</u> angle of attack of the body centerline relative to the local body velocity $\mathbf{U}_p$. The potential expression (6.66) with the strengths (6.72),(6.73), and the geometry given by (6.70) all together

exactly satisfy the 2D Laplace's equation (6.62) and boundary conditions (6.63), (6.64). Therefore this far-field flow happens to be an exact solution of the transverse-flow equations for this geometry.

With the z-doublet strength known, the overall lift can be obtained by integration of the lift gradient (6.69).

$$L \;=\; \int_0^\ell \frac{\mathrm{d}L}{\mathrm{d}x}\,\mathrm{d}x \;=\; \rho V_\infty^2 \int_0^\ell \frac{\mathrm{d}(A\tilde{\alpha})}{\mathrm{d}x}\,\mathrm{d}x \;=\; \rho V_\infty^2\, A(\ell)\,\tilde{\alpha}(\ell) \tag{6.75}$$

This final result shows that within the assumptions of potential flow, the lift of a slender body of revolution depends only on its base area $A(\ell)$ and the local angle of attack $\tilde{\alpha}(\ell)$ at the base location. Trefftz-plane analysis requires that any lifting body must generate trailing vorticity, and indeed the blunt base will shed the necessary vortex sheets as sketched in Figure 6.11. A related conclusion is that bodies which come to a sharp point at the rear cannot generate lift, since they cannot shed vorticity in the absence of flow separation.

The lift distribution $\mathrm{d}L/\mathrm{d}x$ will also produce the following overall pitching moment (positive about the y axis) about the $x=0$ reference point.

$$\mathcal{M}_0 \;=\; \int_0^\ell -x\,\frac{\mathrm{d}L}{\mathrm{d}x}\,\mathrm{d}x \;=\; \rho V_\infty^2 \int_0^\ell -x\,\frac{\mathrm{d}(A\tilde{\alpha})}{\mathrm{d}x}\,\mathrm{d}x \;=\; -\rho V_\infty^2\,\ell\,A(\ell)\,\tilde{\alpha}(\ell) \;+\; \rho V_\infty^2 \int_0^\ell A\tilde{\alpha}\,\mathrm{d}x \tag{6.76}$$

With zero base area and zero camber the lift is zero, and the pitching moment simplifies to

$$\mathcal{M} \;=\; \rho V_\infty^2\,\alpha \int_0^\ell A\,\mathrm{d}x \;=\; \rho V_\infty^2\,\mathcal{V}\,\alpha \tag{6.77}$$

about any reference point x location, where $\mathcal{V} = \int_0^\ell A\,\mathrm{d}x$ is the body's volume. An important result is $\mathrm{d}\mathcal{M}/\mathrm{d}\alpha = \rho V_\infty^2\,\mathcal{V} > 0$. An angle of attack change $\Delta\alpha$ will therefore produce a pitching moment change $\Delta\mathcal{M}$ of the same sign, which will tend to increase $\Delta\alpha$ even more. A body of revolution therefore exhibits an inherent *pitch instability*, which must be stabilized using fins as on a blimp or submarine, or gyroscopically via axial spin as on a finless projectile or a football thrown with a "spiral."

The results above have also been obtained by Ashley and Landahl [50] who used complex mapping to define the transverse flow. They also assumed a more general "finned" body of revolution with local span $b(x)$ whose cross-section is shown in Figure 6.12. The local z–doublet for this case can be given in terms of an effective area A',

$$\kappa_z(x) \;=\; 2V_\infty\,A'\,\tilde{\alpha} \tag{6.78}$$

$$A'(x) \;=\; \pi\left[\frac{R^4}{(b/2)^2} + (b/2)^2 - R^2\right] \tag{6.79}$$

while the local lift relation (6.69) remains the same. The total lift and moment is therefore given by expressions (6.75) and (6.76) if the area A is replaced by the effective area A' defined above. For the case $b/2 = R$ where the fins have zero width, we have $A' = A$, and the finned-body results reduce to the circular cross-section body results as expected.

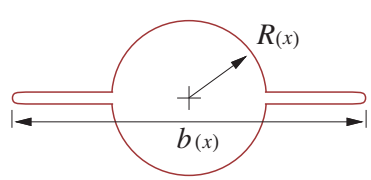

Figure 6.12: Cross section of body of revolution with fins of local span $b(x)$.

6.6.6 Limits of slender-body theory

The conclusion that a rear-pointed body generates no lift assumes strictly potential flow. In reality, slender bodies will have some amount of flow separation if the rear taper is sufficiently rapid or if the local angle of attack is sufficiently large. Fuselages with large rear upsweep angles exhibit such rear separation, and slender delta wings at large angles of attack exhibit *leading edge vortices*. Figure 6.13 shows these flows, and compares them with their strictly-potential flow versions. Although the slender-body approximations and the 2D local far-field expansion (6.66) still remain reasonably valid with such separations, the strengths $\Lambda(x)$ and $\kappa_z(x)$ will become significantly modified. In particular, the free vorticity which is antisymmetric in y will significantly modify the doublet strength κ_z, as can be seen from its definition (2.83). It will therefore modify the local and total lift, and also the moment. Specifically, the separation from the upswept rear body produces a downward bias in the local lift contribution, while the leading edge vortices on a delta wing significantly increase the upward lift, which is known as the *vortex lift* phenomenon. The effect is shown in Figure 6.14 for delta wings of two different aspect ratios.

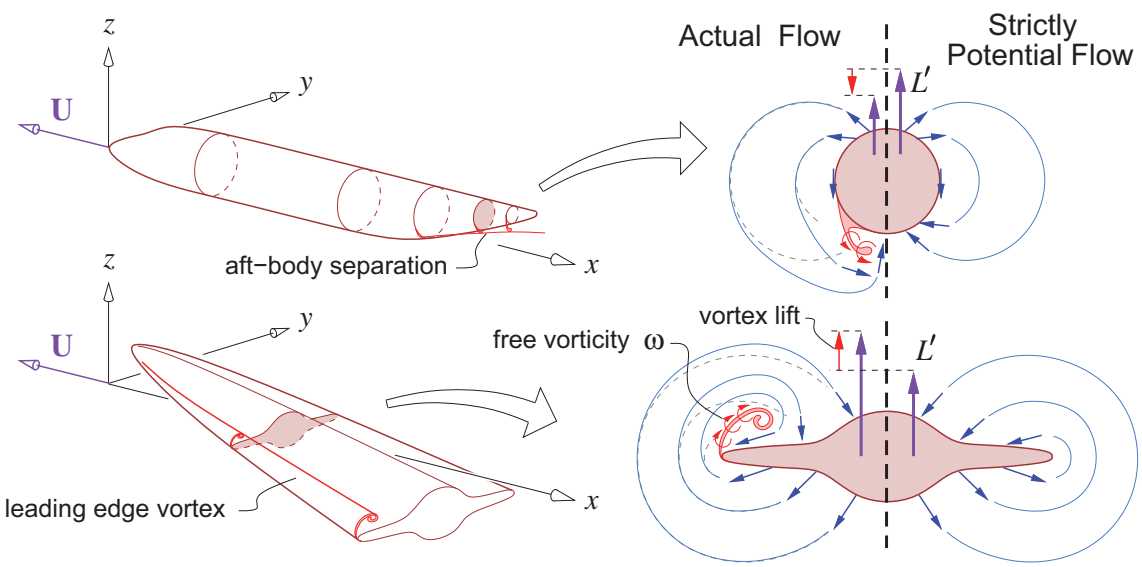

Figure 6.13: Viscous separation from an upswept rear fuselage, and leading edge vortices on a delta wing at high angle of attack. The free vorticity modifies the strictly-potential flow pattern, and also modifies the 2D strengths $\Lambda(x)$ and $\kappa_z(x)$, and the associated aerodynamic loads.

6.6.7 Vortex lift models

Polhamus [51] has developed a method for estimating vortex lift based on the leading-edge suction concept. In this formulation the leading edge suction force magnitude is obtained from the slender-body flow solution for a flate plate, corresponding to the geometry in Figure 6.12 with $R=0$. This force is then applied normal to the surface rather than tangential to the surface, which in effect rotates it by 90° about the leading edge line. The equivalent formulation for general lifting surface geometries uses the leading edge suction force defined by equation (6.23), but applied in the normal direction.

$$\mathbf{F}'_{s_{LE}} = \frac{\pi}{4}\rho C^2\,\hat{\mathbf{n}} \qquad \text{(at sharp leading edge)} \qquad (6.80)$$

This rotation of the leading edge suction force from $\hat{\mathbf{t}}$ to $\hat{\mathbf{n}}$ models the boundary layer vorticity lifting off the surface and forming a free vortex sheet above the surface, as sketched in Figure 6.13 on the bottom right. The force rotation clearly increases the lift, and thus captures the vortex lift contribution. The lift increase reported by Polhamus matches experimental data fairly well.

The rotation of the leading edge suction force changes its direction from partly forward to partly aft, which significantly increases the drag. This effect on drag was also examined by Polhamus [52], with the conclusion that the added drag behaves much like induced drag from the added lift.

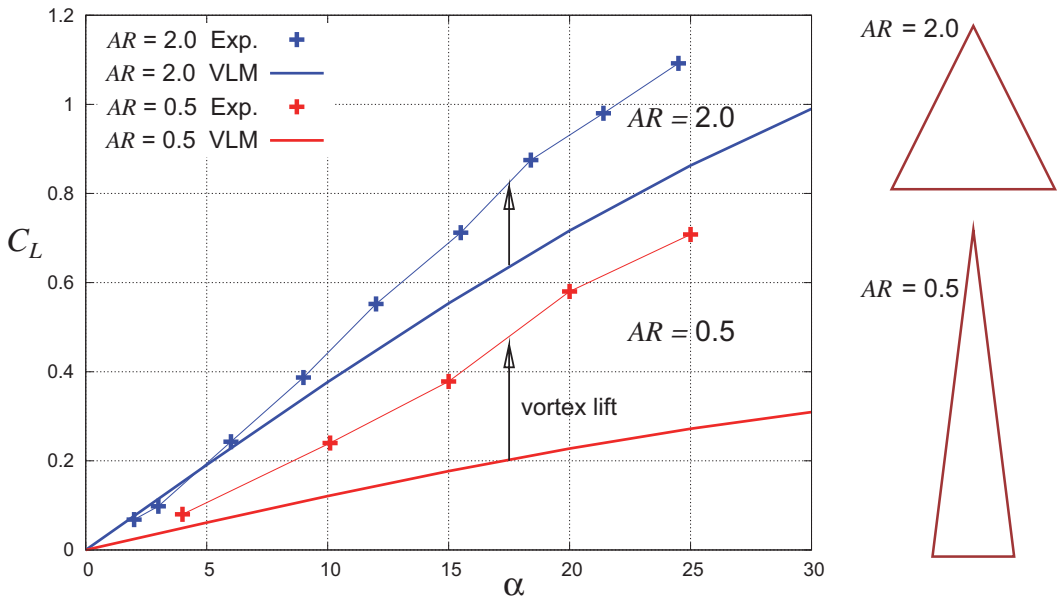

Figure 6.14: Experimental $C_{L(\alpha)}$ for two delta wings from Polhamus [51], compared to Vortex Lattice Method calculations. Excess lift in the experiment at large angles of attack demonstrates the effect of vortex lift resulting from the formation of leading edge vortices, which the standard VLM does not capture. The effect is captured by Polhamus's vortex lift model.